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## Pernod Ricard: Uncorking Digital Transformation

On a gray afternoon in March 2023, Christian Porta, managing director, global business development of French wine and spirits firm Pernod Ricard, sat down in a meeting room in the company's headquarters in Paris, France, which offered expansive views of the city's iconic rooftops. He was waiting for his colleague Pierre-Yves Calloc'h, chief digital officer, who was to update him on the rollout of the company's key digital programs (KDPs).

With more than 90 production locations, operations in over 70 countries, and +20,000 employees as of 2023, Pernod Ricard was the leader in premium international spirits.<sup>1</sup> The company had achieved a leading position in the market, largely through a strategic acquisition strategy and its ability to build a brand and nurture its equity over time. However, pressure to continually add to its extensive brand portfolio to meet discerning customer demands meant that its traditional analog processes could not keep pace, setting Pernod Ricard on a path of digital transformation.

In 2020, Porta and Calloc'h took ownership of a clear direction given by Alexandre Ricard: create a new differentiated capability for the group by using digital tools. Together with the Global Digital Acceleration Team (GDA), they launched the KDPs, a set of programs aimed at using data to optimize processes and make more informed decisions. Two of the main KDPs were D-STAR, a sales recommendation system that provided data-driven insights, and Matrix, a tool that optimized the allocation of advertising spend across brands. The tools were launched in several pilot markets, which started to show successful results.

Nevertheless, Calloc'h, tasked with driving adoption and scalability, encountered varying levels of acceptance across the markets. For example, in France, some managers were reluctant to embrace Matrix due to their emotional attachment to the brands and misalignment of the recommendations with their convictions. Similarly, in Germany, some sales representatives felt that D-STAR was diminishing the strategic element of their job, which hindered the tool's adoption. Furthermore, having to pause the deployment in certain regions due to data limitations, delayed implementation and required additional unforeseen developments and resources.

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Overall, Pernod Ricard needed to address resistance, provide training, align with strategic goals, and overcome data-related hurdles to drive adoption of their digital programs. As they looked out over the Paris rooftops, Porta and Calloc'h pondered the future of the KDPs. Should they launch the KDPs in new markets, despite certain challenges, or focus their resources on scaling them in existing markets?

## History of Pernod Ricard

### *Origins and Early Days*

Pernod Ricard's history began in 1805, when Henri-Louis Pernod founded his distillery, Pernod Fils, in France. Pernod's initial business was the production of absinthe. Absinthe gained popularity as a new offering on the market, and its medicinal properties received a boost when doctors prescribed it to soldiers in the 1840 Algerian War to prevent diseases such as malaria and dysentery. In the 19<sup>th</sup> century, an insect plague destroyed European vineyards, making absinthe a widely available alternative to wine. However, in the early 1900s, the production and consumption of absinthe was prohibited in France and other European countries, following the belief that it caused hyperexcitability and hallucinations. To counter the effects of the loss of business, Pernod Fils partnered with Distillerie Hémard to alter the recipe and create 'Pernod', an anise-based liqueur called pastis in France.<sup>2</sup>

A few years later, Paul Ricard, an entrepreneur and the son of a wine merchant, introduced his own version of pastis, branding it "Ricard," with the vision of creating "the true pastis of Marseille."<sup>3</sup> Calloc'h said, "Ricard's pastis distinguished itself through its unique recipe and Ricard's dedication to quality and authenticity, quickly establishing a devoted following. Ricard became a cultural emblem, encapsulating the essence of Marseille and French conviviality." Pernod Ricard was born in 1975, after the two rival companies merged – according to the press "the equivalent of a merger between General Motors and Ford." The new company solidified its place as a major French spirits company in what became known as the "The Pastis Empire."<sup>4</sup> Its aim was to diversify its product range and to capture international markets.<sup>5</sup>

In 1978, Patrick Ricard, son of Paul Ricard and an astute business man, became chairman and CEO of the group.<sup>6</sup> Under his leadership, Pernod Ricard started an ambitious program of portfolio expansion in the 1980s, including Wild Turkey Bourbon in 1981 (which it later sold to competitor Campari Group to offset the costs of larger strategic acquisitions), non-alcoholic drinks company Compagnie Française des Produits Orangina (CFPO) in 1983 (which it later sold to refocus on spirits), Jameson Irish whiskey in 1988, and leading Australian wine brand Jacob's Creek, in 1989.<sup>7</sup>

In the early 2000s, Pernod Ricard spent close to €20 billion on the acquisition of three well-known wine and spirits producers that it believed had been poorly managed – in 2001 Canadian Seagram Distillers (Chivas, Glenlivet, and Martell), in 2005 British group Allied Domecq (Ballantine's, Malibu, Mumm, Perrier-Jouët), and in 2008 Vin & Sprit (Absolut) from the Swedish Government.<sup>8</sup> According to Porta, these acquisitions were "the three big bangs that catapulted the company from number seven in the world of spirits to number two." In 2009, Pernod Ricard made "Créateurs de convivialité" its new corporate signature, committing to promote responsible conviviality and hosting.<sup>9</sup> Patrick Ricard passed away in 2012, and his nephew, Alexandre Ricard, became CEO of the company in 2015.<sup>10</sup>

By this time, further large acquisition targets had dried up. Porta said, "There were two or three other companies that were of interest to us, but they were not for sale, for example the U.S. whisky brand Jack Daniels." Pernod Ricard turned its attention to organic growth and bolt-on acquisitions of small, craft brands as its growth strategy. In 2016, the group purchased a majority stake in Black Forest Distillers GmbH, followed in 2017 by a majority stake in Smooth Ambler (a bourbon producer) and Del

Maguey Single Village (the number one mezcal in the United States), and, in 2019, acquired Malfy, Rabbit Hole Whiskey, Castle Brands and Firestone & Robertson Distilling Co.<sup>11</sup>

### *The Group in 2023*

As of 2023, Pernod Ricard was the leader in premium international spirits operating more than 90 production sites, and employing almost 19,500 people.<sup>12</sup> With a market capitalization of €55 billion in 2023, it was headquartered and listed in Paris, and part of the CAC 40 and Eurostoxx 50 indices.<sup>13</sup> Bonds were Pernod Ricard's most important source of financing, accounting for 90% of gross debt. It had net debt/EBITDA of 2.4x at the end of fiscal 2022.<sup>14</sup>

The group's strategy was set on four main pillars: premiumization, product innovation, digital acceleration, and active portfolio management.<sup>15</sup> Pernod Ricard's brands were divided into five brand categories (see **Exhibit 1**), and the company held one of the largest portfolios of premium brands in the industry – both international and local.<sup>16</sup> In early 2020, Pernod Ricard's CEO, Alexandre Ricard, reflected, "By 2030, there will be 2 billion more middle-class people in this world. People are drinking less but drinking better, and we expect our strong position in luxury spirits to be one of the factors accelerating our growth."<sup>17</sup> In 2022, Pernod Ricard's marketing expenses were estimated at about €1.7 billion – approximately 16% of sales.<sup>18</sup>

In 2022 Pernod Ricard reported revenue of €10.7 billion (see **Exhibit 2** and **Exhibit 3**). Its key market was the Asia Pacific region, where it generated the majority of its sales (41%). Europe and America represented about 30% of sales each.<sup>19</sup> Analysts believed Pernod Ricard's financial performance benefited largely from its pricing strategy, which was driven by its diverse and balanced premium portfolio and the lifting of Covid-19-related restrictions. Pernod Ricard's growth level exceeded pre-Covid-19 levels in all its operating regions, despite still expecting some short-term disruption in China. Calloc'h said, "In a context of unprecedented inflation on cost of goods and in view of preserving margins, agile revenue growth management, starting with pricing, was essential."

Pernod Ricard's main competitor was the British multinational alcoholic beverage company Diageo, founded in 1997. In 2022, it reported revenue of €17.4 billion.<sup>20</sup> Diageo owned over 200 brands, including Johnnie Walker, Guinness, Tanqueray, Baileys, Smirnoff, Captain Morgan, Don Julio, Ciroc, and Buchanan's.<sup>21</sup> Other important competitors included the U.S. beer, tequila, and vodka producer Constellation Brands, and the world's largest brewer, the Belgian firm Anheuser-Busch InBev.

### *Organizational Structure*

In 2023, Pernod Ricard's brands were produced in over 90 sites across the world and distributed in over 160 countries, (**Exhibit 4**). The group had over 70 affiliates, each with its own local team and P&L statement.

Pernod Ricard had historically been a decentralized organization to fit the nature of its business, and to cater to varying consumer tastes and legal constraints around alcohol. The company consisted of a global headquarter in France, autonomous affiliates, brand companies, and market companies across the globe (see **Exhibit 5**).<sup>22</sup> Brand companies focused on the development of brand strategy and activations, and market companies implemented the overarching strategy locally. Whilst the overall direction and management stemmed from the global headquarters (HQ) in Paris, the group respected the operational autonomy of the market and brand companies. The decentralized structure aimed to provide managers with increased flexibility in adapting to local specificities.<sup>23</sup>

## Pernod Ricard's Digital Transformation Journey

A number of catalysts set Pernod Ricard on the path of digital transformation. According to a report by McKinsey & Company, in 2018, digital transformation efforts had accelerated in all industries, with 46% of organizations reporting the implementation of digital strategies, up from 23% in 2017.<sup>24</sup> For example, Diageo implemented digital tools already in 2017. The company launched Catalyst, a marketing analytics tool that merged various data sources to determine the appropriate budget for each brand in the portfolio.<sup>25</sup> Such developments pressured Pernod Ricard to re-evaluate its own digital journey. This was particularly highlighted when investors referred to competitors' tools, questioning the pace of Pernod Ricard's work on digitalization.

Another catalyst was the complexity of managing a growing brand portfolio with traditional processes and structures. Between 2015 and 2018 alone, Pernod Ricard acquired 13 brands.<sup>26</sup> While HQ managed the brand acquisitions, the management responsibilities were transferred to the market companies, which struggled to engage with audiences to increase brand awareness efficiently. With increasing consumer fragmentation in the industry, there was pressure to constantly add to the group's offering. Porta explained, "Consumers want to have a repertoire of drinks – a drink for the week, a drink for the weekend, a glass of wine, a cocktail. We therefore need to have a bigger portfolio. If you have six blockbuster brands, it's relatively easy for the human brain to handle on its own. But if you have 13 blockbusters, 15 craft brands, plus additional strategic local brands, it becomes increasingly complex."

Of concern, also, was that some of Pernod Ricard's markets, like France, were mature and faced stagnating sales. Furthermore, whilst holding a leading position in the retail and off-trade channels (liquor stores, supermarkets), the company struggled to be the leader in the on-trade channel (bars and restaurants), leading some senior executives to call for digitalization as a possible solution.

In 2018, the group developed a strategic plan, *Transform and Accelerate*, that contributed to Pernod Ricard's aim of growing between 4-7% per year in net organic sales. The plan outlined 21 inflection points for the period 2018-2021, with many focused on the digital transformation of the business. Porta said, "While we saw numerous things as being important to reach our strategic goals in an increasingly complex world, we centered our plan on how to become more data driven and efficient in the way we spend our money." Pernod Ricard's plan had three dimensions: organization, business processes, and cultural mindsets.

### *Gaining Momentum*

In 2019, Pernod Ricard evaluated that the program was working well, but was not accelerating fast enough, and hired the Boston Consulting Group (BCG) to accelerate the digital transformation. Building on Alexandre Ricard's guidance to create data and artificial intelligence (AI) capability, the digital team and BCG interviewed over 100 senior executives at the company to map the needs of the company, as well as where the impact would be the biggest.

Across the group, one of the cultural shifts was to push HQ to work closer with the affiliates. Etienne Coulon, head of product for D-STAR, who had previously worked for a Pernod Ricard regional development team, recalled, "Traditionally, new projects stemmed from the affiliates, and, if successful, were rolled out across the group. There was general reluctance towards pitches and inputs from HQ that risked being inappropriate for local markets. Under the new digital strategy, we aimed to use the tools globally and harmonize data governance, and we could not risk excessive local adjustments. We therefore globalized the creation of the digital programs at HQ-level."

Against this backdrop, the company began to redefine decentralization. Porta described, “Decentralization means that we provide sales and marketing employees with the best global tools for them to make the best possible choices. It is not: ‘I can do whatever I want in any situation.’ But rather: ‘I have the best tools and data at my disposal, which allows me to make the best decision efficiently.’”

The goal of Pernod Ricard’s digital journey was to equip the company for four battlegrounds: win in key markets, build passion brands, fund the journey, and value its people.<sup>27</sup> Supported by BCG, in 2020, Pernod Ricard appointed its first Chief Digital Officer, Calloc’h, and created the digital acceleration team. Calloc’h was tasked with making the digital changes concrete for the company’s business. “It was not solely about data science capabilities, it was about transformation, of which data science constituted an important pillar,” described Jean-William Cousin, digital acceleration office director in Calloc’h’s team. The goal of the transformation was to make technology the new competitive advantage, allowing for the anticipation of market trends and increased speed and agility in decision making.<sup>28</sup> It was spearheaded by the development of four KDPs, a suite of data applications and tools, based on data and predictive AI, developed in-house.<sup>29</sup>

## Key Digital Programs

In early 2020, the digital acceleration team considered the big potential digital use cases that would be relevant for the spirits industry and, specifically, Pernod Ricard. Ideas included supply chain and bottling optimization, as well as social selling via influencers. Calloc’h explained, “We mapped the potential benefits versus costs of implementation. Since the margin on our products is vast, we saw that supply chain projects would bring far less value than marketing and sales projects.”

In April 2020, after a full day of reviewing the 20 selected projects, Pernod Ricard’s executive committee prioritized six. Four of them became core business KDPs and two became new business ventures. The four core KDPs were: Matrix, a Marketing-Mix Modeling (MMM)<sup>a</sup> program that optimized resource allocation across advertising channels;<sup>30</sup> D-STAR, which aided sales representatives by optimizing planning and providing action recommendations based on geographical, demographic, and business data;<sup>31</sup> Vista Rev-up, which determined optimal promotions by simulating future scenarios; and Maestria which employed AI and data to predict consumer choices.<sup>32</sup> Of these, D-STAR and Matrix had matured the fastest.

From the outset, Pernod Ricard opted to develop its KDP solutions in-house, to retain full control, instead of using what was available on the market, and to build a competitive advantage. Calloc’h explained the reasoning, “Off the shelf solutions would need a lot of tweaking with high recoding costs, to meet our required level of granularity and flexibility. For example, our three-tier market in the United States differs significantly from the United Kingdom. In addition, it’s difficult to find data scientists who want to gain expertise in these specific environments. Also, our tools contain many strategic decisions. For each KDP, we have about 50 business parameters, which we wouldn’t like to communicate to a third party.”

The global COVID-19 pandemic and related lockdowns gave the KDPs an unexpected boost. Porta explained, “It highlighted the impact of technology and the importance of the digital projects we had already put in place. The executive committee understood that things are changing and that we need to move fast.” COVID-19 also helped us recruit people for our digital team who were available because

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<sup>a</sup> Marketing mix modeling is a statistical analysis technique used by marketers to measure and evaluate the impact of various marketing variables on sales or other key performance metrics (KPIs). The marketing variables most often include price, product, promotion, and place.

other projects had dried up. That was also the case with our BCG consultants. We probably got the best on the market because so many other companies were freezing consulting projects.”

## D-STAR

Prior to D-STAR, sales representatives used to plan their visits by relying on fixed frequency rules, simple classification of outlets, and their own heuristics. D-STAR leveraged data and machine learning algorithms to provide sales teams with data driven insights and recommendations. Its objective was to allow sales teams to optimize business planning and prioritize the next best action in sales outlets, such as determining which potential clients to target or which products to promote, in order to maximize sales and profitability (**Exhibit 6**).

From a technical perspective, the tool clustered stores with 30 to 50 sources of data, e.g. size of store, brands sold, demographics, sales of Pernod Ricard products, and proximity to points of interest such as a beach or barbecue spots. The sales performance of each store was then compared to others in its cluster. Each Monday morning, sales representatives opened the D-STAR application and received a list of recommended stores to visit during the week according to their potential for sales uplift. They were presented with a set of actions to be taken, tailored to each store visit. Typical actions included negotiating a new stock-keeping unit (SKU), increasing order size, or implementing a new promotion.

Based on the type of recommendation provided, sales outlets were categorized into two groups: nuggets and diamonds. Nuggets referred to outlets where the algorithm recommended a higher number of visits compared to before. Diamonds were outlets where the recommendation was very different from the usual actions taken. Some stores might show little potential when considered in isolation, but in combination with data on social demographics and neighborhood amenities, the algorithm re-segmented the outlet universe and proposed actions for sales uplift.

Internal modelling of algorithms based on data availability and local specificities allowed the D-STAR to adapt to the varying routes to market that Pernod Ricard faced in its different geographies. There was no common code base for all of D-STAR. Coulon said, “The application of a D-STAR use case or feature in market A versus market B could be rather different. There is no way we can fully copy paste or simply reuse the code. The idea is to keep some sort of consistency in the features that we can deliver to focus the efforts of the central teams in charge. For example, imagine you provide a private chef with ingredients to make you whatever you want. We wanted to shift from that approach to a restaurant where you can select what you want based on a set menu. Similarly, D-STAR provides a catalogue with set features for each market to choose from.”

In order to function correctly, D-STAR required robust data (see **Exhibit 7** for data requirements). For markets where there was insufficient data to implement D-STAR, like in smaller markets, such as the Netherlands or Belgium, Pernod Ricard developed a less demanding version – D-STAR Light. “Either we thought D-STAR AI was possible, or we used D-STAR Light. We never used both versions in the same market,” said Cousin.

Each deployment was expected to last a minimum of 9 months, and to involve a local deployment team of at least 10 people (see **Exhibit 8** for composition of local deployment team).

### *Pilot Markets for D-STAR*

In November 2020, D-STAR was piloted in four markets: France, Germany, the United States (in the state of Florida), and India (two states). Coulon commented, “We selected these markets, because we

wanted to test if D-STAR would stick in different route to market environments and test data collected in different environments. For example, although France and Germany could be regarded as similar market environments, Pernod Ricard France was able to access more retail customer data.”

To deploy D-STAR in the pilot markets, HQ needed to convince the markets to change their ways of working. The D-STAR Product team implemented test, learn, and optimize (TLO) periods of at least four to six months, in order to measure the impact of the tool on sales teams’ performance. It did not impose the use of the recommendations, preferring sales reps to continue being the masters of their environment. However, the team tracked to what extent recommendations were followed.

Ioana Morogan, chief transformation officer at Pernod Ricard France, outlined how the TLO approach was beneficial. “We started with a small group of three to four key test users, who worked daily with the data and change management team. We made iterative improvements and the initial users became ambassadors once D-STAR was ready to be scaled.” Initial users also had a hotline to call when encountering issues. The TLO approach also allowed the team to identify potential missteps in the implementation process when seeking user buy-in. Morogan recalled: “We started to implement the tool without discussing the ‘why’ and the ‘what’s in it for me.’ We were driven by making the tool operational and not the communication, which is key. If people don’t understand why the tool is being introduced when using it, and how it will help them make smarter, fact-based decisions, they will not use it.”

The TLO approach also allowed Pernod Ricard to realize how it might need to follow a different approach across markets to secure buy-in. In France, for example, it was important to have management buy-in, which cascaded to the sales representatives, who were inclined to follow the direction provided by the sales directors. In contrast, in Germany, the approach had to be more decentralized, as field sales staff acted as business development executives and had more responsibilities. The focus there was to onboard individuals who were usually resistant to change, rather than relying on those more open to new systems. Wolf Geiling-Rasmus, Head of Key Account Management for Pernod Ricard Germany, recalled, “We involved those who were reluctant earlier than others, to give them the chance to get participate and to show that their feedback was valuable.” According to Coulon, Germany initially constituted one of the most challenging experiences. He explained, “Sales representatives in Germany expressed more doubt and initial skepticism, fearing that the recommendations would take away from their commercial acumen and ability to choose what to do.” Additionally, the tool was not initially embedded into their existing sales software. Geiling-Rasmus, said, “Sales reps had to do twice the effort. They had to work within the regular system, whilst also doing the planning in a separate system.”

Key feedback received from the initial testers of D-STAR in France was that it helped them gain a more complete understanding of the portfolio and its needs. Additionally, in stores where a new visit frequency was applied (recommended by D-STAR), the net market share growth was +0.26 distribution points, and net sales growth was +1.2% compared to the control group. In Germany, after surmounting the initial challenges, and better integrating D-STAR into the existing workflow, the adoption rate of D-STAR recommendations by Pernod Ricard sales teams increased to 85%. Incentives played a role: if the reps met at least 60% of the tool’s proposals, they received an additional bonus. Clara Thomas, transformation & strategic projects director for Pernod Ricard Western Europe, said, “However, if staff had not felt that the tool improved their work and increased their sales, they would not have used it.”

In the United States, a key success factor had been that D-STAR was promoted to sales reps as a competitive advantage. Jack Frohlinger, senior director of commercial strategy and transformation at Pernod Ricard USA, said, “When we emphasized that the insights from D-STAR would give sales reps an advantage in the marketplace compared to competitors, the adoption rate significantly improved.”

## Matrix

Matrix provided data and AI powered recommendations on how to optimize the budget allocation across brands in a given market, and across media channels, such as digital, TV, social media, and in-store brand activation (**Exhibit 9**). While marketing spend had an impact on long-term brand building, Matrix as a tool could not directly calculate the long-term effect, and thus additional parameters were used to account for it. Sarah Nicolas, marketing effectiveness manager for France, recalled, “Prior to Matrix, many decisions related to marketing spending were made partly based on intuition, due to the absence of tools capable of guiding them and measuring ROI.” Matrix allowed marketers to make more informed decisions. In turn, they had to argue their case if they did not follow the recommendations.

From a technical point of view, the algorithm used a minimum set of data in a country, consisting of the past three years of weekly sales data and compared it to the marketing activities carried out. Calloc’h said, “We tried with monthly sales data and with only one or two years of data, but it wasn’t effective. We eventually decided on the required minimum in order to optimize. However, the level of data granularity required was certainly a challenge.” The algorithm created a response curve that showed the expected impact based on the level of investment, based on which it recommended how to invest in the future. In comparison to D-STAR, where the code base was tailored for each market to a much larger extent, Matrix was composed of 80% of core code and 20% of market-specific code.

Matrix users needed to have a certain skill set to get the most out of the tool. The ideal profile had a statistical brain, knowledge of the market, and business acumen. Calloc’h expanded, “When something doesn’t seem right, they should be able to determine whether it’s a data issue, an algorithm issue or a market insight. 80% of the time, it’s a data problem; 10% of the time, it’s an algorithm problem; and 10% of the time, it’s an insight. For example, we got the unusual recommendation to push champagne promotions in summer and not Christmas. During Christmas, people will buy our brand anyway, whilst in summer, there are fewer promotions, and so it is a good opportunity to remind people of our champagnes.”

Furthermore, Matrix marked a change in dialogue between the marketing teams, the finance and sales teams. Calloc’h said, “Previously, marketing teams had to constantly justify its spending to the finance team. Now, with Matrix analyzing and predicting sales impact, the finance team could advocate for increased investment in successful campaigns.”

### *Pilot Markets for Matrix*

In October 2020, Pernod Ricard launched Matrix in two pilot markets: Germany and Japan. The rationale was to test it in two mid-sized markets. “If you fail, you would not want to fail in too large a market, to avoid jeopardizing large amounts of investments,” stated Antoine Genot, global marketing effectiveness director. It was also important that the pilots were replicable to other markets. For example, Germany had replicability to Western Europe, and Japan had replicability in a number of countries in Asia. A further reason for selecting Japan was that it was rich in on-trade channels, such as bars and restaurants, and not solely focused on off-trade.

As with D-STAR, the deployment of Matrix in the pilot markets was through a TLO phase. To ensure that Matrix was running correctly and to measure its impact, the data team compared budgets optimized by Matrix with a control budget without Matrix. Genot described, “We wanted to trace the ROI evolution of the Matrix budget versus the control budget, to confirm that Matrix was working efficiently. However, it took a year, i.e. two data refreshes, before the impact could be observed. Once Matrix began to generate positive results in Germany and Japan, it validated the success of the KDP.”



Running the two pilot projects in parallel provided a continuity buffer when issues emerged. For example, the deployment of Matrix in Japan had to be delayed due to difficulties in acquiring sales data. At first, the deployment team believed that data with the required weekly granularity was non-existent. Yet, after investigation, they discovered that in many instances it had simply not been digitized, and the project was paused for a few weeks in order to rectify the issue. “If we had launched Matrix solely in Japan without seeing success in the second pilot country, perhaps we would not have had the confidence to continue the project,” stated Calloc’h. Japan provided a learning experience; for any subsequent deployment in other markets, a data team would go to the market three months prior to do all the checks. After three months, the full team would enter to deploy the program.

In the case of Germany, the deployment team included a German-speaking deployment manager from HQ. “His ability to exchange with local employees without language barriers was of great importance,” said Geiling-Rasmus. The deployment team was also composed of an employee responsible for the local helpline and collecting feedback. He would transmit any questions and feedback to HQ, to clarify any issues.

The deployment of Matrix required a larger mindset and workflow change from users compared to D-STAR, given the underlying creative nature of the marketing profession. Morogan said, “the difference between D-STAR and Matrix is that for D-STAR, the defining user is the sales representative, who needs to execute and amplify a predetermined strategy without needing to understand the algorithm’s mechanics. In contrast, in Matrix, if the brand director, who used to take strategical decisions in a certain way, finds the recommendation incomprehensible, it’s more problematic. This highlights the need for upskilling all marketers, including basics of statistical analysis.” The purpose of the marketing team in Pernod Ricard was “to create emotion.” With the arrival of Matrix, this slogan was challenged, with marketeers being urged to also become more analytical and more structured when working with data and tracking.

## Scaling the Key Digital Programs

In early 2021, with the KDP pilots demonstrating success, it was time to consider how the initiatives could be scaled, from a technical, cultural, and organizational perspective.

From a technical perspective, the KDP team had to first adjust the technology, since in the pilots it took shortcuts to deliver fast results. Calloc’h said, “Now we have to move from ‘quick and dirty’ to a more thorough approach. We need to rewrite almost all the algorithms because they were made to prove the case.” Calloc’h’s team also had to find the right balance between standardization and adaptation. Genot said, “The beauty of internalizing the tool was that we could cater to local specificities, but we also need programs that are transferable from one market to another for them to be scalable. We can’t create a tailor-made solution for each market.” Another obstacle to widespread deployment was data availability. Calloc’h said, “The data exists in all markets, but to get it is complex. Some retailers don’t want to share their sales data, for instance.”

From a cultural perspective, a key issue was to ensure continuous utilization of the tools. Porta said, “There needs to be adoption from the organization, not only from individuals. Marketing people move from one brand to another, they come in, they come out. That’s a struggle for the continuous adoption of Matrix. There is more stability in sales teams than in marketing teams.” It was also key to strike the right balance between users relying on the tools for high-quality recommendations, and HQ’s questioning them if they did not make sense. Porta said, “The sales and marketing teams cannot decide to not use the tools only because of personal expectations.” Responsibility for faulty recommendations was also a concern. Whilst recommendations should be followed as a rule of thumb, individuals could

reject them for legitimate reasons. Coulon added, “If the algorithm is providing a list of stores and actions, but sales teams are dropping them, we will investigate whether the issue lies with the algorithm or deviation by the user and the reasons behind it.” As during the pilots, recommendations were filtered through a “control tower,” to sort elements that the algorithm was unaware of, such as temporary ‘out-of-stocks’ or vacation plans of store managers.

From an organizational perspective, BCG was instrumental in getting Pernod Ricard’s digital transformation off the ground, by providing the technical competency and know-how. However, from the start, the key ambition for Pernod Ricard was to create a competitive advantage. Cousin explained, “If you are following a partner, however strategic that partner is, you are not creating competitive advantage. We aimed to create internal capabilities, but had no clue how to do that. BCG helped us understand how things are done, and a next step was to figure out if and how we could continue on our own. Which roles would we need? What kind of data science organization would we need? Would it jeopardize the growth of the KDPs? We were breaking all the barriers imaginable in a larger group like Pernod Ricard, so all options were on the table.”

### *Developing a Data Science Organization at HQ*

In March 2021, Pernod Ricard started to gradually internalize all operations and take over from BCG. Calloc’h, together with David Lepicier, global data and analytics director, had prepared for the transition and started creating a small internal team. They organized tech recruitment nights, where they recruited, in a few hours, 8-10 new staff, mainly data scientists and engineers. In addition, the recruitment of a number of freelancers accelerated the transition, bringing the team to 15 additional staff members. The recruitments resulted in a newly formed Global Digital Acceleration team (GDA) (see **Exhibit 10**).

Lepicier had a clear idea of the profile for the new data organization. He explained, “We don’t want to give too specific orders, we want to empower people to do things themselves. I will not challenge my data scientist on the right data modeling, but rather give clear directions aligned with our business objectives and ensure that there are efficient interactions among colleagues.” Furthermore, Lepicier wanted to replicate BCG’s agile approach to the GDA. Compared to the more traditional and pyramid-shaped organizational structure of Pernod Ricard’s marketing and sales teams, the GDA had a network organization design, with a flatter management scheme (see **Exhibit 11**). Team members working on the KDPs were assigned cross-functional squads for day-to-day activities, but remained under the GDA umbrella. Lepicier said, “The reporting is not as interesting to me as how teams deliver. Lead data scientists are shaping programs and staff managers are shaping teams, the combination of both makes us highly efficient.” Throughout 2021 and 2022, the GDA team grew to 130 data staff, which represented about 10-15 new recruits per month.

Lepicier was preoccupied with continuity. He said, “We want people to stay with us for a long time. I have only two main goals as a leader: our programs need to deliver impactful result for our markets, and our people should have a good time doing it.” Nevertheless, attempting to keep up with the demand for data scientists, Lepicier recruited 50% of new staff on freelance contracts. Replacing half of the initial BCG consultants with external freelancers raised concerns of them leaving hastily. “In fact, this was unfounded,” said Cousin. “We saw brilliant engagement from the freelancers, who were eager to build a long-term relationship.” Furthermore, by employing freelancers, Pernod Ricard could recruit very specific profiles, which was important given the project-based nature of the GDA’s activities. If all freelancers had been hired on Pernod Ricard contracts, managing their long-term career development within the company, especially for project-specific roles, would have been challenging.

As the team grew, Lepicier realized that he could not continue managing them directly. He recalled, “At some point, I had 26 people reporting to me. It required 100% of my time.” Thus, he suggested to hire a ‘staff manager’ for the data team, yet finding the right candidate was more difficult than expected. Lepicier recalled, “When I explained the job, many interviewees were either tech experts who wanted to code rather than develop people, or managers who wanted to focus on program steering rather than ensuring that teams are performing effectively.”

Building a data science organization at HQ-level resulted in a much stronger and more centralized organization in the context of the KDPs. Thomas said, “The “what” came very much from HQ, whilst the market companies were on the receiving end, and dealt mainly with the “how” around it.” The business knowledge and vision were led mainly by the HQ business department. It made decisions on the future business capabilities of the group and held the product ownership role. All the transversal capabilities required to execute that vision were managed in the GDA, led by Calloc’h.

### *Expanding in Growth Markets*

By March 2023, Pernod Ricard started scaling the KDPs in new growth markets: D-STAR was rolled out in the United Kingdom, China, South Africa, and Canada, while Matrix in France, China, U.S., Spain, United Kingdom, Germany, Canada, Korea, Brazil, South Africa, Japan, and Mexico.

Initially, the KDPs were globally steered by the GDA team, but also embarked business and tech teams, both at HQ level and in markets. At market level, the coordination was managed by transformation offices to foster coherence. Thus, each KDP deployment brought together a team of people from various backgrounds. Calloc’h described, “Each team includes various profiles, such as a data analyst, data engineer, machine learning engineer, someone from finance, someone from marketing, a project manager, and a person in charge of training.” Upon entering the scaling phase, the initial deployment team left, and there was a shift from project-focus to product-focus, to ensure the longevity of the KDPs. This shift led to the creation of the role of product owners, who were tasked with managing the continuous maintenance and deployment of the tools. Coulon said, “The product teams are composed of a product owner, a deployment manager, and a senior data scientist, who together constitute the product trio. The product owner glues all the tasks and responsibilities together, and ensures good communication.” The goal of the product team was to be the intersection between the data teams and the business side of the KDPs.

As the scaling continued, some affiliates anticipated that when they would reach maturity, the KDPs would no longer fall under the responsibility of the digital transformation office, but instead be incorporated into the day-to-day business. Colin Kavanagh, chief transformation officer for North America, said, “D-STAR is a great example. By the end of FY25, we’ll probably move local responsibility out of the Transformation Office and put into the local day-to-day business. Because we’ve scaled it up, we’ve built that capability, it will be part of the way of working of the commercial team. It’s no longer transformation, it’s part of what they do.”

Furthermore, while during the pilots HQ did not demand the markets to deliver more, in the roll-out phase, as the deployment accelerated, the KDPs were integrated into the market companies’ budgets and strategic three-year plans. Calloc’h said, “If a project is expected to generate an additional \$1 million in profit, HQ asks markets to allocate at least 60% of it to their targets. This way, if they deliver 100%, they would overperform their targets. These projects are challenging for the affiliates due to significant shifts in work methods and the acquisition of large amounts of data. By incorporating the commitment into their balance sheets, they are incentivized to fully engage with the projects.”

Nonetheless, the adoption rates of the new tools across markets were much higher for D-STAR (85%) than Matrix (60-70% on average). Matrix's recommendations often required difficult decisions, such as cutting certain brands or touchpoints, leading to resistance among marketing directors accustomed to traditional methods and media. Nicolas said, "Marketeers don't necessarily like that a data-driven tool says that action A is more efficient than action B, especially when the results are different from what they expect. They need to change their point of view, their habits, but without knowing at the beginning if it will be the right decision." Particularly in some markets, the marketing team faced issues with the reluctance of managers to overcome their emotional attachment to brands and enforce Matrix's recommendations. For instance, some marketing directors demonstrated resistance to cutting investments on brands that they had been managing historically, or that had a higher internal status, while being more open to reducing budgets on others. Such resistance, driven by emotions, was particularly challenging to navigate. Finance teams in certain countries were also skeptical about the ability of the digital technologies to calculate the impact of marketing on the P&L, which slowed down the adoption of these new tools.

To address these challenges, the KDP team organized two educational programs: one for all employees and one for top management. Employee training sessions, conducted individually and in small groups, emphasized that AI would not replace human decision-making. The program for top management, called Platform Pioneer, was held by the change management team every six to eight weeks. Calloc'h said, "It is crucial to communicate what we do – how the KDPs work, what doesn't work, why. Education is really key." The executive committee was also updated monthly with presentations and app demonstrations by users.

Comparing the implementation of Matrix in the United States and China, different dynamics were at play. In the United States, the implementation of Matrix coincided with a change in leadership and a transition to a performance-based mindset. As a result, the adoption of Matrix in the region was high. Mary Beth Jowers, vice president for portfolio strategy at Pernod Ricard USA, said, "The KDPs came at a perfect time to enable the overall transition. Matrix is just one of the aspects that fed this transition to focusing on proven performance KPIs and continuous improvement." In China, the hierarchical structure and played a significant role in driving adoption. Yann Soenen, vice president marketing at Pernod Ricard China, described, "Role models are even more important in some cultures. It was important for me to demonstrate interest in the tool, in order to foster adoption throughout the rest of the team."

The gradual integration of Matrix into decision-making processes, such as quarterly investment sign-offs and yearly brand plans, fostered a change in work culture and further solidified adoption. However, limitations in readily available data from different channels and the lack of detailed historical investment tracking in certain areas, such as e-commerce, presented ongoing challenges in leveraging Matrix effectively.

Management was conscious that Pernod Ricard needed to be sensitive in how it set the rate of change in the organization. Nicolas said, "The group is changing so rapidly, and the KDPs are challenging everything in a very fast way. There are always people who find it unsettling and difficult to adapt to change, and you won't affect every person in the same way. The question is to find the right balance in terms of not overloading the organization with change."

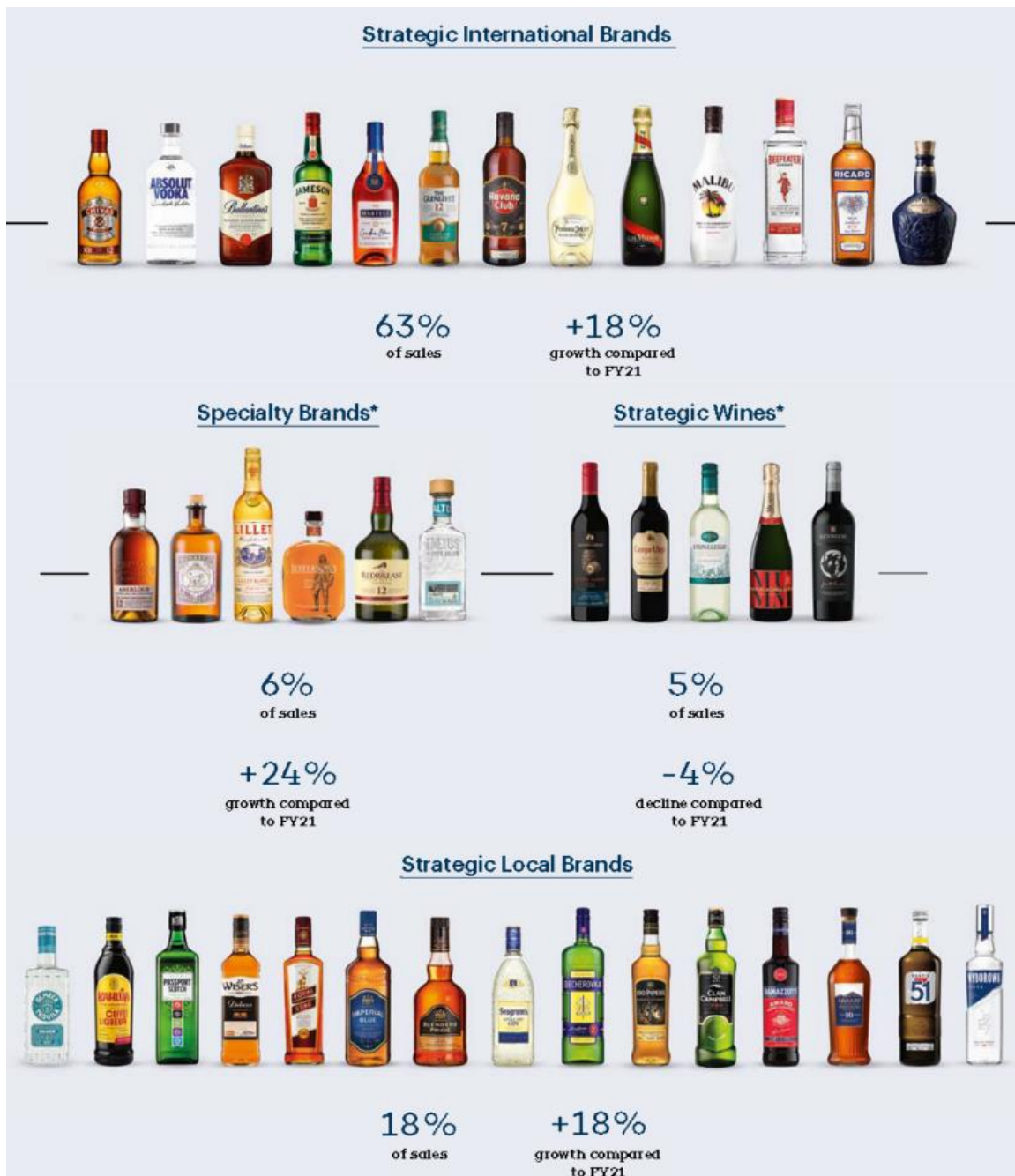
## The Way Forward

In the meeting room, Porta and Calloc'h contemplated the future steps of the KDPs at Pernod Ricard. As of March 2023, the KDPs had proven their value through increase of sales between 1.5% and

4.5% depending on markets deployed, but also an increase of up to 15% on marketing efficiency, while new ways of making decisions were increasingly deployed across the Group. Porta commented, “We want to take things further and faster, on the tools that we have decided on and designed, but also find what are the right next use cases.” They wanted to build on the momentum that they had constructed, but needed to address the challenges faced in driving adoption, especially for Matrix, where common resistance to change and emotional attachment to traditional marketing methods posed significant obstacles. To reinforce adoption of the Matrix tool required substantial time and effort for training and education, high level managerial support, and an understanding of the specific dynamics at play in each of its markets.

As Porta and Calloc'h gazed over the Parisian rooftops, they faced important decisions about the future of the KDPs, and in particular Matrix. Pernod Ricard needed to find a way to address resistance, provide effective training and support, align with strategic goals, and overcome logistical and data-related hurdles. They wished to expand Matrix further into new markets, but could they do this in parallel to reinforcing adoption where Matrix had already been launched? The future direction of the KDPs also depended on answering the question: should they focus on continuing the expansion into new markets, or should they focus on further scaling inside existing ones? The decisions made would shape the path forward for Pernod Ricard and its key digital programs.

Exhibit 1 Pernod Ricard's Selection of Brands



Source: "Annual financial report, FY 22," Pernod Ricard, 2023, <https://www.pernod-ricard.com/sites/default/files/2022-03/Sustainability-Linked%20Financing%20Framework%20final.pdf>, accessed February 2023.

**Exhibit 2** Pernod Ricard Annual Consolidated Income Statement, FY 2022

| € million                                               | 30.06.2021   | 30.06.2022   |
|---------------------------------------------------------|--------------|--------------|
| Net sales                                               | 8,824        | 10,701       |
| Cost of sales                                           | (3,531)      | (4,228)      |
| <b>Gross margin after logistics expenses</b>            | <b>5,293</b> | <b>6,473</b> |
| Advertising and promotion expenses                      | (1,393)      | (1,698)      |
| <b>Contribution after advertising and promotion</b>     | <b>3,900</b> | <b>4,775</b> |
| Structure costs                                         | (1,477)      | (1,751)      |
| <b>Profit from recurring operations</b>                 | <b>2,423</b> | <b>3,024</b> |
| Other operating income/(expenses)                       | (62)         | (62)         |
| <b>Operating profit</b>                                 | <b>2,361</b> | <b>2,963</b> |
| Financial expenses                                      | (410)        | (308)        |
| Financial income                                        | 39           | 48           |
| <b>Financial income/(expense)</b>                       | <b>(371)</b> | <b>(260)</b> |
| Corporate income tax                                    | (667)        | (676)        |
| Share of net profit/(loss) of associates                | (4)          | 5            |
| Net profit of discontinued and held for sale activities | 0            | -            |
| <b>Net profit</b>                                       | <b>1,318</b> | <b>2,031</b> |
| o/w:                                                    | -            | -            |
| • non-controlling interests                             | 13           | 35           |
| • attributable to owners of the parent                  | 1,305        | 1,996        |
| Earnings per share – basic (in euros)                   | 5.00         | 7.71         |
| Earnings per share – diluted (in euros)                 | 4.99         | 7.69         |

Source: “Annual consolidated financial statements, FY 22,” Pernod Ricard, 2023, <https://www.pernod-ricard.com/sites/default/files/inline-files/10.%20Consolidated%20financial%20statements%20FY22.pdf>, accessed February 2023

**Exhibit 3** Pernod Ricard Consolidated Balance Sheet, FY2022**Assets**

| € million                                         | 30.06.2021    | 30.06.2022    |
|---------------------------------------------------|---------------|---------------|
| Net amounts                                       |               |               |
| <b>Non-current assets</b>                         |               |               |
| Intangible assets                                 | 10,725        | 11,512        |
| Goodwill                                          | 5,505         | 6,145         |
| Property, plant and equipment                     | 3,177         | 3,591         |
| Non-current financial assets                      | 685           | 761           |
| Investments in associates                         | 36            | 243           |
| Non-current derivative instruments                | 65            | 4             |
| Deferred tax assets                               | 1,623         | 1,844         |
| <b>Total non-current assets</b>                   | <b>21,816</b> | <b>24,100</b> |
| <b>Current assets</b>                             |               |               |
| Inventories and work in progress                  | 6,555         | 7,369         |
| Trade receivables and other operating receivables | 1,126         | 1,388         |
| Income taxes receivable                           | 141           | 145           |
| Other current assets                              | 413           | 435           |
| Current derivative instruments                    | 8             | 32            |
| Cash and cash equivalents                         | 2,078         | 2,527         |
| <b>Total current assets</b>                       | <b>10,321</b> | <b>11,896</b> |
| Assets held for sale                              | 11            | 15            |
| <b>Total assets</b>                               | <b>32,147</b> | <b>36,012</b> |



## Liabilities

| € million                                                     | 30.06.2021    | 30.06.2022    |
|---------------------------------------------------------------|---------------|---------------|
| <b>Shareholders' equity</b>                                   |               |               |
| Capital                                                       | 406           | 400           |
| Share premium                                                 | 3,052         | 3,052         |
| Retained earnings and translation differences                 | 10,066        | 10,496        |
| Group share of net profit                                     | 1,305         | 1,996         |
| <b>Group shareholders' equity</b>                             | <b>14,829</b> | <b>15,944</b> |
| Non-controlling interests                                     | 246           | 309           |
| <b>Total shareholders' equity</b>                             | <b>15,075</b> | <b>16,253</b> |
| <b>Non-current liabilities</b>                                |               |               |
| Non-current provisions                                        | 253           | 318           |
| Provisions for pensions and other long-term employee benefits | 477           | 361           |
| Deferred tax liabilities                                      | 2,825         | 3,139         |
| Bonds – non-current                                           | 8,787         | 9,238         |
| Non-current lease liabilities                                 | 405           | 400           |
| Other non-current financial liabilities                       | 108           | 179           |
| Non-current derivative instruments                            | 0             | 18            |
| <b>Total non-current liabilities</b>                          | <b>12,854</b> | <b>13,653</b> |
| <b>Current liabilities</b>                                    |               |               |
| Current provisions                                            | 163           | 150           |
| Trade payables                                                | 2,337         | 3,019         |
| Income tax payable                                            | 282           | 263           |
| Other current liabilities                                     | 1,134         | 1,311         |
| Bonds – current                                               | 70            | 842           |
| Current lease liabilities                                     | 103           | 107           |
| Other current financial liabilities                           | 122           | 406           |
| Current derivative instruments                                | 6             | 9             |
| <b>Total current liabilities</b>                              | <b>4,218</b>  | <b>6,107</b>  |
| Liabilities related to assets held for sale                   | 0             | 0             |
| <b>Total liabilities and shareholders' equity</b>             | <b>32,147</b> | <b>36,012</b> |

Source: “Annual consolidated financial statements, FY 22,” Pernod Ricard, 2023, <https://www.pernod-ricard.com/sites/default/files/inline-files/10.%20Consolidated%20financial%20statements%20FY22.pdf>, accessed February 2023

Exhibit 4    Pernod Ricard’s Global Presence with Local Manufacturing Units



Source: “Annual financial report, FY 22,” Pernod Ricard, 2023 <https://www.pernod-ricard.com/sites/default/files/2022-03/Sustainability-Linked%20Financing%20Framework%20final.pdf>, accessed February 2023

Exhibit 5    Pernod Ricard’s Brand and Market Companies

| BRAND COMPANIES            | PERNOD RICARD HEADQUARTERS | MARKET COMPANIES <sup>2</sup>     |
|----------------------------|----------------------------|-----------------------------------|
| THE ABSOLUT COMPANY        |                            | NORTH AMERICA                     |
| CHIVAS BROTHERS            |                            | PERNOD RICARD ASIA                |
| MARTELL MUMM PERRIER JOUET |                            | EMEA & LATAM <sup>3</sup>         |
| IRISH DISTILLERS           |                            | GLOBAL TRAVEL RETAIL              |
| PERNOD RICARD WINEMAKERS   |                            | PERNOD RICARD FRANCE <sup>4</sup> |
| HAVANA CLUB INTERNATIONAL  |                            |                                   |

Source: “Annual financial report, FY 22,” Pernod Ricard, 2023 <https://www.pernod-ricard.com/sites/default/files/2022-03/Sustainability-Linked%20Financing%20Framework%20final.pdf>, accessed February 2023

Exhibit 6 Screenshot of D-STAR

**Weekly D-STAR Recommended visits**

**Schedule Drag & Drop**

**Completed Weekly planning**

**D-STAR Recommended visits**

| Rang | Code      | Semaine | Nom                       |
|------|-----------|---------|---------------------------|
| 5    | 740120850 | 39      | GEANT 74100 ANNEMASSI     |
| 7    | 730081674 | 39      | GEANT 73100 AIX LES BAINS |
| 9    | 740260001 | 39      | CASINO SM 74330 BALME     |
| 10   | 730650020 | 39      | LECLERC 73000 CHAMBERY    |
| 11   | 731810010 | 39      | CASINO SM 73600 MOUTRIER  |
| 12   | 730060928 | 39      | LECLERC 73210 AIME        |
| 13   | 740109735 | 39      | MONOPRIX 74012 ANNECY     |
| 14   | 740100330 | 39      | MONOPRIX 74000 ANNECY     |

**Drag & Drop**

**Drag & Drop**

Source: Company documents.

## Exhibit 7 D-STAR Data Requirements

## DSTAR – Data Requirements

| Data Domain    | Data family            | Data set                                                                  | Data example                                                                        |
|----------------|------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Product        | Product Definition     | • Name, description, brand, brand quality, size                           | • Absolut Blue 70cl, ABSOLUT, ABSOLUT BLUE 0.7                                      |
|                | Product Categorization | • Product category                                                        | • VODKA                                                                             |
|                | Product ID             | • Product ID: SKU, EAN, ERP code                                          | • Absolut Blue 70cl regular bottle, 93720172634527, ABS-BLUE-0.7                    |
| Trade customer | Customer profile       | • Customer name<br>• Banner name<br>• Banner channel<br>• POS Unique code | • Walmart<br>• Walmart Hypermarket Boston<br>• Channel type<br>• 23104              |
| Sell-out       | Retail                 | • PR Sell out<br>• Competition sell out                                   | • Sell out (Volume, Value), dates (weekly), SKU, BQS, POS<br>• Value, volume, dates |
| Sell-in        | Customer Orders        | • Internal Net Sales Value                                                | • Absolut Blue 70cl – Walmart – 15\$                                                |
| Sales Visits   | N/A                    | • Sales reps names<br>• Store visited<br>• Visit date                     | • John Doe<br>• Walmart Hypermarket Boston<br>• 01/02/2022                          |

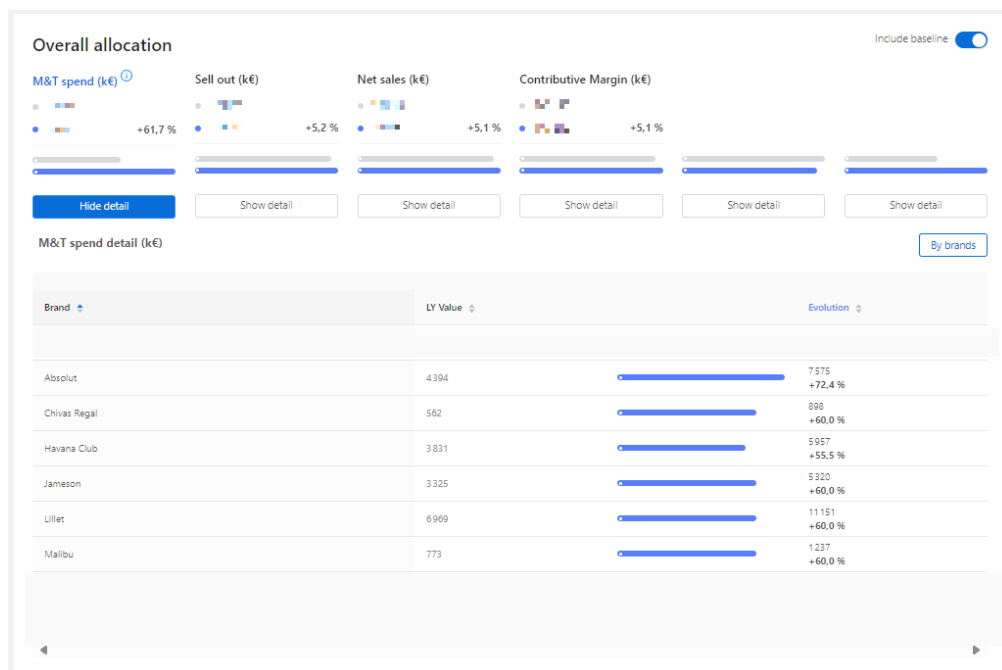
Source: Company documents.

## Exhibit 8 Local Resources Required to Deploy D-STAR

| Team            | Role                                   | #                                            | %    | Duration   | Comments                                                      |
|-----------------|----------------------------------------|----------------------------------------------|------|------------|---------------------------------------------------------------|
| Local resources | KDP Sponsor                            | [1 – 3]                                      | 10%  | 9 months   | Local CODI member(s)                                          |
|                 | KDP Program Owner                      | 1                                            | 10%  | 9 months   | Head of transformation or similar                             |
|                 | KDP Project Manager                    | 1                                            | 50%  | 9 months   | Local business project manager                                |
|                 | IT Project Manager                     | 1                                            | 50%  | 9 months   | Local / regional IT/Data manager                              |
|                 | Data Engineer                          | 1                                            | 100% | 9 months   | Local / regional Data managers                                |
|                 | Business planner                       | 1                                            | 40%  | 9 months   | Could be the same profile                                     |
|                 | SFA Admin                              | 1                                            | 10%  | On going   |                                                               |
|                 | KDP Champion                           | 1                                            | 20%  | 6 months   |                                                               |
|                 | Field Sales team members / Super users | 20% of the field sales team (incl. Managers) | 20%  | 4-6 months | During TLO/test phase + hypecare                              |
|                 | Local Data Steward / Data coordinator  | 1                                            | 20%  | Recurring  | Key focus on Data pre-assessment, data preparation & run mode |

Source: Company documents.

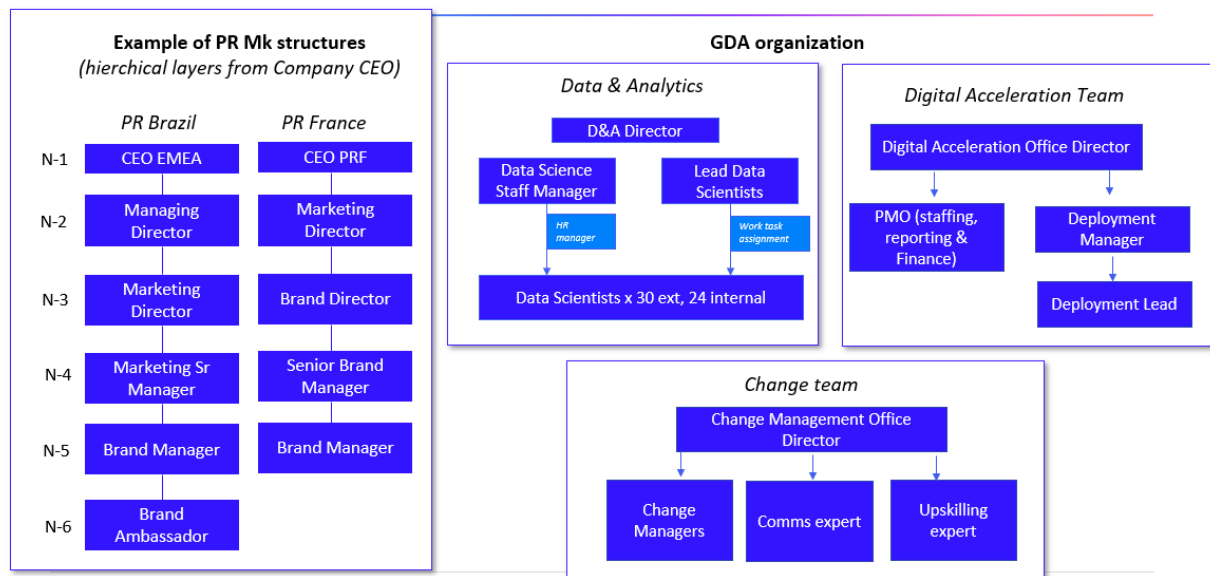
Exhibit 9 Screenshot of Matrix



## Response Curve



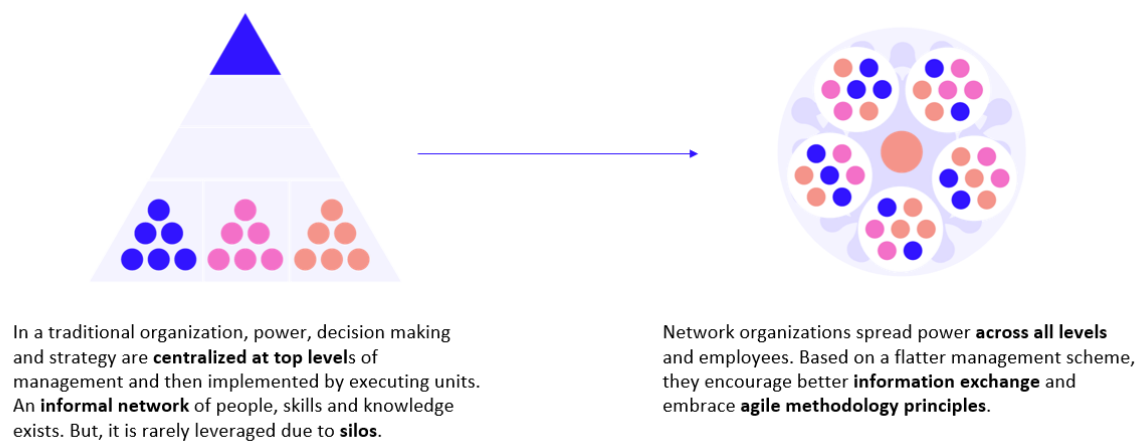
Source: Company documents.

**Exhibit 10** Pernod Ricard Global Data Acceleration Team Organization

Source: Company documents.

**Exhibit 11** Pernod Ricard's Global Digital Acceleration (GDA) Team Organization Structure

From traditional pyramidal and siloed organizations to network organizations



Source: Company documents.

## Endnotes

- <sup>1</sup> “The Pernod Ricard Market View”, based on IWSR volume data at end 2022.
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- <sup>11</sup> Pernod Ricard, “Our History,” Pernod Ricard website, <https://www.pernod-ricard.com/en/our-group/our-history>, accessed in February 2023.
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- <sup>14</sup> Richard Withagen, “Pernod Ricard,” Equity Research from Kepler Cheuvreux, February 1, 2023, via Refinitiv, accessed in February 2023.
- <sup>15</sup> Danny Yeo Sze Wai, “Pernod Ricard SA - Stock Report,” CFRA, February 4, 2023, accessed via Capital IQ, accessed in February 2023.
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